Soda Gun Project

Advanced Methods in Mechanical Design

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1. Clarification of the problem

1.1 Description of the company

FN S.r.l. is a leading manufacturer known for its high-quality beverage dispensers, including soda guns, ice makers, and post-mix systems. FN S.r.l. focuses on innovative solutions that enhance beverage dispensing efficiency and hygiene. FN S.r.l. design its products with an emphasis on reliability, ease of use, and maintenance, often featuring customizable options to meet specific user needs and always prioritizing food safety standards, ergonomic designs, and user-friendly interfaces to ensure optimal performance in various commercial settings. FN S.r.l primarily caters to commercial customers in the hospitality industry, including bars, restaurants, cafes, and hotels. These businesses require reliable, efficient, and easy-to-maintain beverage dispensing systems to serve large volumes of drinks quickly and consistently.

For example, bars and nightclubs need high-capacity, durable soda guns that can handle frequent use and dispense a variety of beverages rapidly. Features like easy customization, ergonomic design, and visibility in low-light conditions are crucial for these environments.

Restaurants and cafes value versatility and reliability. They require systems that can dispense a range of drinks, including sodas, sparkling water, and juices, with ease of use and maintenance being paramount. These systems must handle peak times efficiently without compromising performance. Hotels and resorts often offer a wide variety of beverages to guests throughout the day, necessitating systems that are both functional and aesthetically pleasing, with customization options to match different themes and settings.

Event venues and catering services need portable and robust systems that can be set up and operated with minimal hassle. Quick setup, intuitive interfaces, and dependable performance are critical for these customers who often deal with large gatherings. Corporate offices and facilities may also install high-end beverage dispensers in their cafeterias or break rooms, prioritizing ease of use, variety in drink options, and low maintenance.

There is also a growing market for sophisticated beverage dispensers in high-end home kitchens. Home enthusiasts look for top-tier, customizable systems that offer a range of drink options and integrate seamlessly with smart home technology. Common needs across all these customer types include reliability and durability to ensure consistent performance, ease of use with intuitive interfaces and ergonomic designs, and customizability for personalized settings and branding. Maintenance and cleanliness are also crucial, with systems designed for easy cleaning to uphold hygiene standards, and sleek, modern designs enhance the visual appeal of the service area.

1.2 State of the art

This state-of-the-art soda mixing and dispensing system combines advanced technology, ergonomic design, and high-quality materials to provide a superior user experience. It's designed for professional environments like bars and restaurants, as well as home use, with a focus on ease of use, customization, and maintenance. The system is both aesthetically pleasing and highly functional, ensuring it stands out in the market.

Overall, this state-of-the-art design not only addresses the functional requirements of a soda dispensing system but also elevates the user experience through technological advancements and thoughtful design, positioning it as a competitive and attractive option in the market.

COMPONENTS OVERVIEW AND USER INTERFACE

- **Smart Soda Gun**
  - **Material**: High-quality, food-grade stainless steel and durable ABS plastic with a rubberized, ergonomic grip.
  - **Display**: OLED touchscreen for intuitive control and user feedback.
  - **Lighting**: Integrated RGB LED lights for visibility and aesthetic appeal.
  - **Connectivity**: Wireless communication with the basement unit via Bluetooth or Wi-Fi.
  - **Nozzle**: Precision multi-channel nozzle for accurate mixing and anti-drip.
  - **Feedback**: Haptic feedback for user interaction confirmation.
- **Basement Unit**
  - **Material**: Brushed stainless steel with customizable panels.
  - **Mounting**: Versatile mounting options for bar counters, walls, or under-counter installation.
  - **Cooling System**: Advanced thermoelectric cooling for efficient and quiet operation.
  - **Pumps**: High-precision peristaltic pumps for accurate dosing and smooth flow.
  - **User Interface**: Touchscreen control panel for system management and diagnostics.

FUNCTIONAL REQUIREMENTS

- **Mixing and Dosing**
  - **Technology**: Microcontroller-based precise control over syrup, water, and CO_2 mixing.
  - **Customization**: User-programmable settings for various drink recipes, including different carbonation levels.
  - **Sensors**: Flow sensors and level indicators for real-time monitoring and adjustments.
- **Flow Control**
  - **Components**: High-quality solenoid valves for precise control.
  - **Sensors**: Flow meters for consistent output without spills.
  - **Piping**: Food-grade silicone tubing for durability and safety.
- **Refrigeration**
  - **System**: Thermoelectric cooling elements with fan assistance.
  - **Temperature Control**: Digital thermostat for precise temperature settings.
- **Sparkling Water Integration**
  - **CO_2 System**: High-efficiency carbonator with adjustable settings for different carbonation levels.
- **Basement Control Interface**

- **Display**: High-resolution touchscreen for system status, ingredient levels, and maintenance alerts.
- **Integration**: Smart home compatibility for remote monitoring and control.
- **Alerts**: Visual and audio notifications for low levels, maintenance needs, and errors.

ERGONOMICS AND USABILITY

- **Design**
  - **Ambidextrous**: Ergonomic design for both right and left-handed users.
  - **Grip**: Rubberized handle for comfort and non-slip usage.
  - **Weight**: Lightweight design for ease of use.
- **Visibility**
  - **Lighting**: RGB LEDs around the nozzle and handle for easy location in low light.
  - **Colour Coding**: Customizable colour settings for different drinks and notifications.

SAFETY AND MAINTENANCE

- **Food Safety**
  - **Materials**: All components in contact with beverages made from food-grade materials.
  - **Design**: Smooth, easy-to-clean surfaces to prevent bacterial growth.
- **Maintenance**
  - **Cleaning**: Modular design for easy disassembly and cleaning.
  - **Alerts**: Scheduled maintenance alerts via the control interface and mobile app.

AESTETHICS AND CUSTOMIZATION

- **Appearance**
  - **Design**: Sleek, modern look with customizable panels and finishes.
  - **Lighting**: Customizable RGB lighting for brand alignment or personal preference.
- **Customization**
  - **Branding**: Options for logos and custom colours.
  - **Interface**: Customizable interface with different language options and drink profiles.

USER EXPERIENCE

- **Quick Setup**: Guided setup process with on-screen instructions.
- **Intuitive Operation**: Simple and intuitive interface for selecting and dispensing drinks.
- **Maintenance Alerts**: Automated alerts for ingredient levels and required maintenance.
- **Custom Recipes**: Ability to create, save, and recall custom drink recipes.

1.3 List of Requirements

External Property	Internal Properties	Target Values
Performance and functionalities		

	pour different drinks	from 5 to 12 on the market
	pour just soda, still water or sparkling water	
	pour fluid with different temperature	room temperature, cold
	regular flow	see state of the art
	able to be integrated in a hydraulic circuit	
	able to contain and move fluid	pressure on state of the art
	able to be programmed	set different drinks and their order on the selection interface
	able to be installed in commercial service shop and in private houses	
	able to be placed in many positions of the workstation (on a plane or fixed at the wall)	
	easily grabbed and carried	
	easily found under low light conditions	
	easily and daily cleaned and ordinarily maintained	
	able to communicate to the user the levels of the ingredients and some warnings	
	able to input/save/find information about the kind of ingredients and the wanted mixing.	
	Fast and precise	
Ergonomics		
	able to be programmed easily by the user	This operation should be done by user without hydraulic and electrical knowledge
	easy to be grabbed and carried several times per hour	

	lightweighted and limited in volume	Less than 2 kg
	easy to be understood	a shape known in our culture (like a handgun shape)
	easily and daily cleaned and ordinarily maintained	
	easy to find in the dark	
	able to communicate to the user the levels of the ingredients and some warnings	
	easy to be used by both right and left-handed users	
	easy to be used by inexperienced bartender	
Reliability		
	avoid fluid losses	
Maintenance		
	easily and daily cleaned and ordinarily maintained	
Safety		
	avoid fluid losses	
	food compatible	
	avoid dangerous contacts between fluid and electrical conductors	
	avoid injuries during usage	
Aesthetics		
	customable	
	gorgeous, fashioning	
	lights equipped	
	good colours and shapes	
Fitness to standards and laws		

	food compatible	
	avoid dangerous contacts between fluid and electrical conductors	
Fitness to assembly/disassembly		
	easily and daily cleaned and ordinarily maintained	
Costs		
	competitive in the market	under 500 euro

1.3 Time scheduling

	25-04	25-05	25-06	16-07
Clarification of the problem				
Concept Design				
Preliminary Lay-out				
Executive Design				

Tab 1 – Gantt Diagram

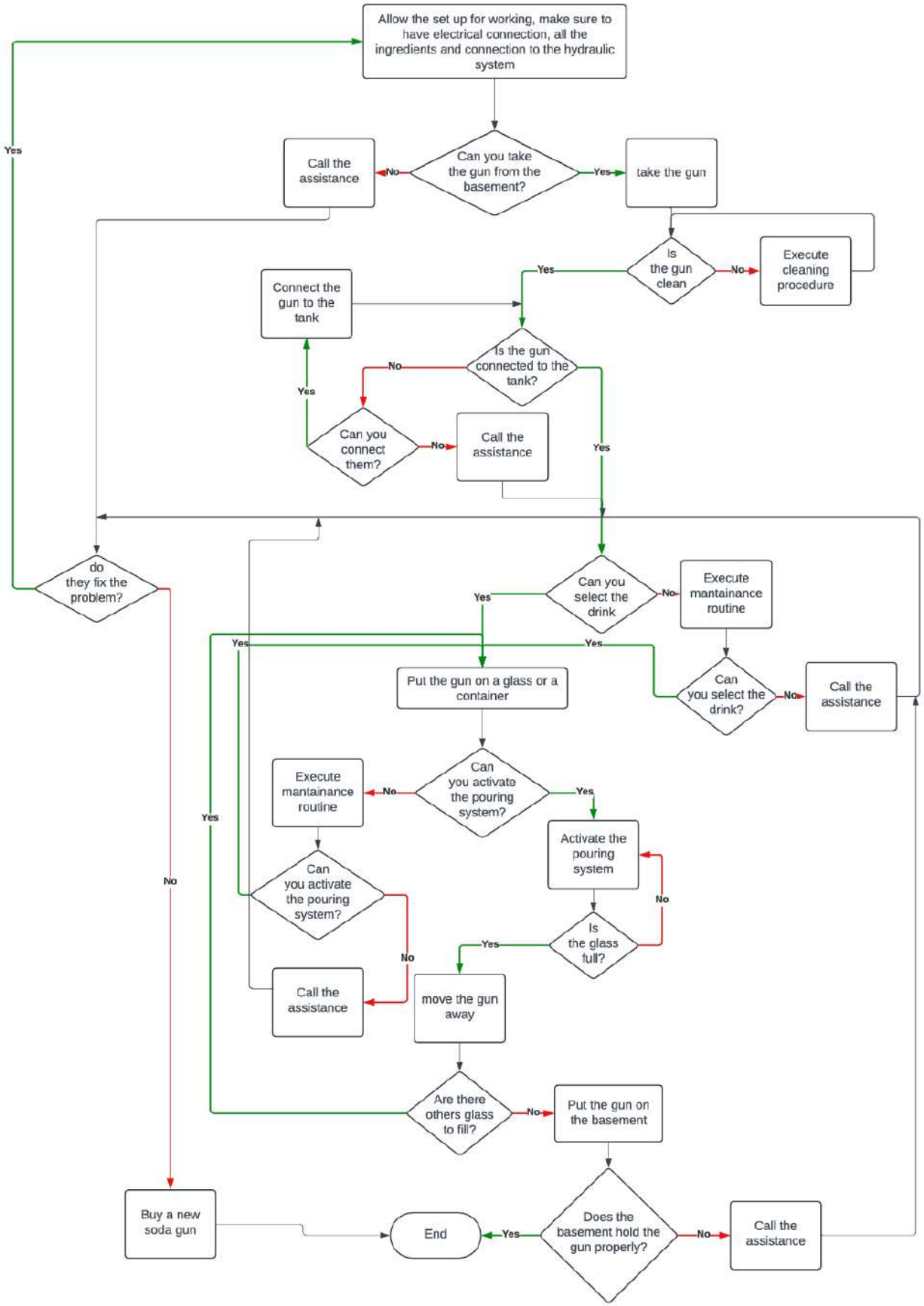
2. Concept Design

2.1 Sequential Functional Analysis

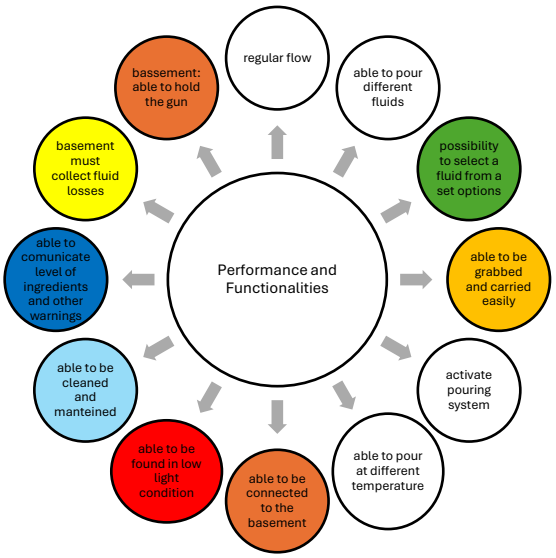
In the first moment we cannot define the two routines, which are strictly related to the design and the assembly of the soda gun. In SFA we use the cleaning routine and the maintenance routine as a function, but they should be well defined, also considering the possible inexperience of the user.

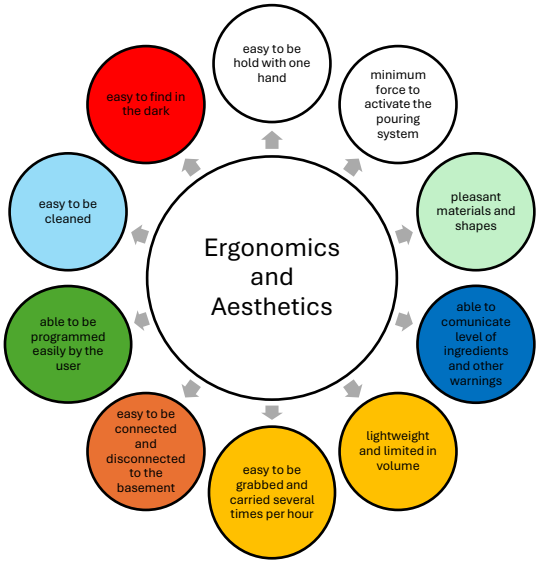
These two operations require the complete disassembly of the soda gun to facilitate the removal of sugar accumulations or other substances that can clog the tubes or create interference, preventing the relative motion of some parts.

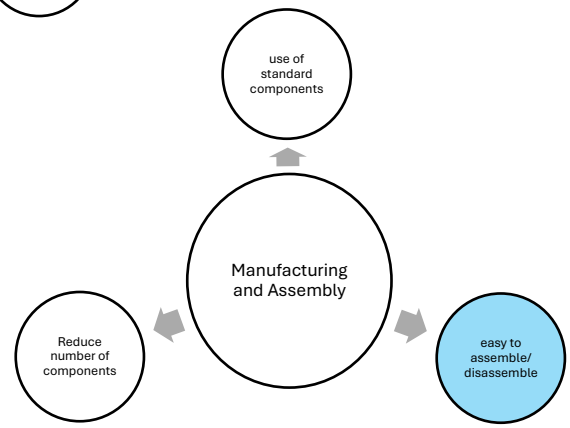
Whenever one of these two operations don't allow start to work again, assistance must be called. The service team will, if possible, restore the product to optimal working conditions. Otherwise, it will be necessary to replace the product with a new one. This flowchart guides the user in detail on how to use the soda gun, using a flow or block diagram with events and decision gates.

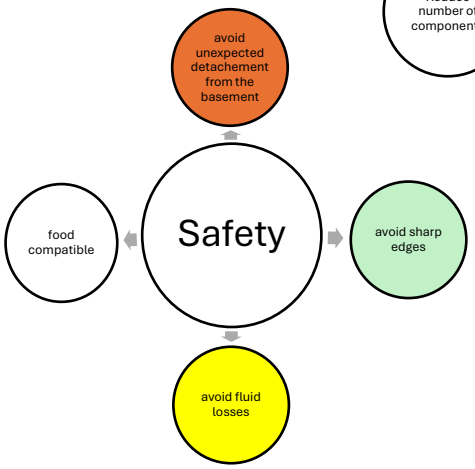

Allow the set up for working, make sure to have electrical connection, all the ingredients and connection to the hydraulic system
Can you take the gun from the basement?
Yes
No
Call the assistance
take the gun
Is the gun clean
Yes
No
Execute cleaning procedure
Connect the gun to the tank
Is the gun connected to the tank?
No
Yes
Can you connect them?
No
Yes
Call the assistance
do they fix the problem?
Yes
No
Can you select the drink
Yes
No
Execute mantainance routine
Can you select the drink?
Yes
No
Call the assistance
Put the gun on a glass or a container
Can you activate the pouring system?
No
Yes
Execute mantainance routine
Activate the pouring system
Can you activate the pouring system?
No
Is the glass full?
No
Yes
Call the assistance
move the gun away
Are there others glass to fill?
Yes
No
Put the gun on the basement
Does the basement hold the gun properly?
Yes
No
Call the assistance
Buy a new soda gun
End

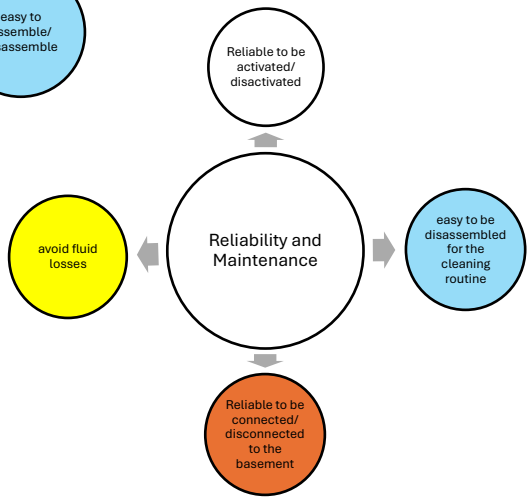
2.2 Tree Functional Analysis





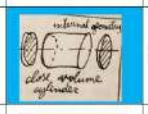
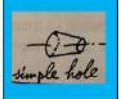
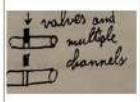
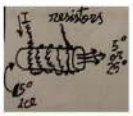
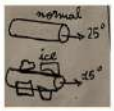
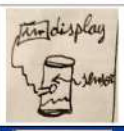


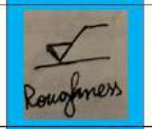




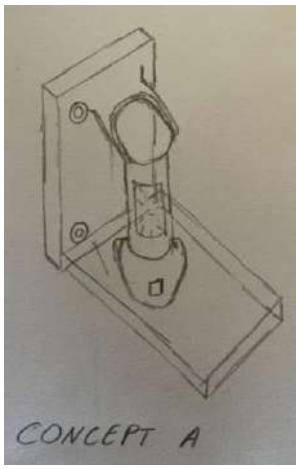
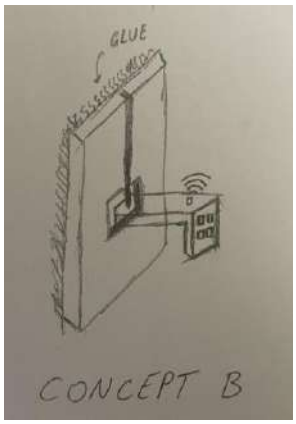
lights on the gun must be delivered both in performance and ergonomics	cleaning process involves performances, functionalities, ergonomics, aesthetics, manufacturing, assembly, reliability and maintenance	warnings control is not only a good point in terms of ergonomics but it increases performances as well	an ease of use in term of programmability make the soda gun funtional and ergonomics
avoid fluid losses should be accomplished both for safety and reliability and maintenance and functionalities of the basement	low weight increase performances in terms of possibility to use the gun for a long time without troubles, making it ergonomics too	A clamped basement ensure reliability, safety, good performances and ease of use of the soda gun	smooth edges help in both safety and ergonomics and aesthetics

2.3 Morphological Matrix

			Erg	S-R	Aes	Cos		Erg	S-R	Aes	Cos		Erg	S-R	Aes	Cos		Erg	S-R	Aes	Cos
1	to contain fluid		5	5	4	5		5	4	4	5		5	3	5	4					
2	to move fluids		5	5	5	2		3	3	5	3		5	5	5	4					
3	to pour fluids		4	4	5	4		5	3	5	5		5	5	4	2					
4	to select different drinks		5	5	4	3		4	5	3	3		3	2	3	4					
5	to pour different drinks		4	3	3	4		4	4	3	3		5	5	2	1					
6	to pour at different temperatures		4	3	3	4		4	4	3	2		5	4	3	2					
7	to be connected with the basement		4	3	4	2		2	3	3	4		5	4	5	3					
8	to be found with low light		5	4	5	3		5	5	5	4		3	2	3	5		5	3	4	5
9	able to be cleaned		5	5	5	2		4	4	5	3		4	3	4	4					
10	to communicate fluids level		4	3	5	5		4	3	4	3		3	2	5	5	NONE	5	5	2	5
11	to input/save/find info about drinks		4	4	2	2		4	4	3	5	NONE	5	5	2	1					
12	avoid loss of fluid		5	5	5	2		4	4	4	2		3	2	3	2					
13	usable with one hand		5	5	4	3		3	3	5	5		4	4	4	3					

14	able to be assebled/ disassembled		4	5	3	4		4	3	4	3		3	4	3	4					
15	avoid slipping		5	4	5	2		5	3	5	3		4	2	4	3					
16	explain how using		3	3	3	3		4	3	2	3												
17	basement must be connected with a fixed surface		3	4	3	4	glue	4	2	3	3	weight-friction	4	3	3	3					
	CONCEPT A	261																			
	CONCEPT B	252																			

2.4 Concept aggregation and sketches

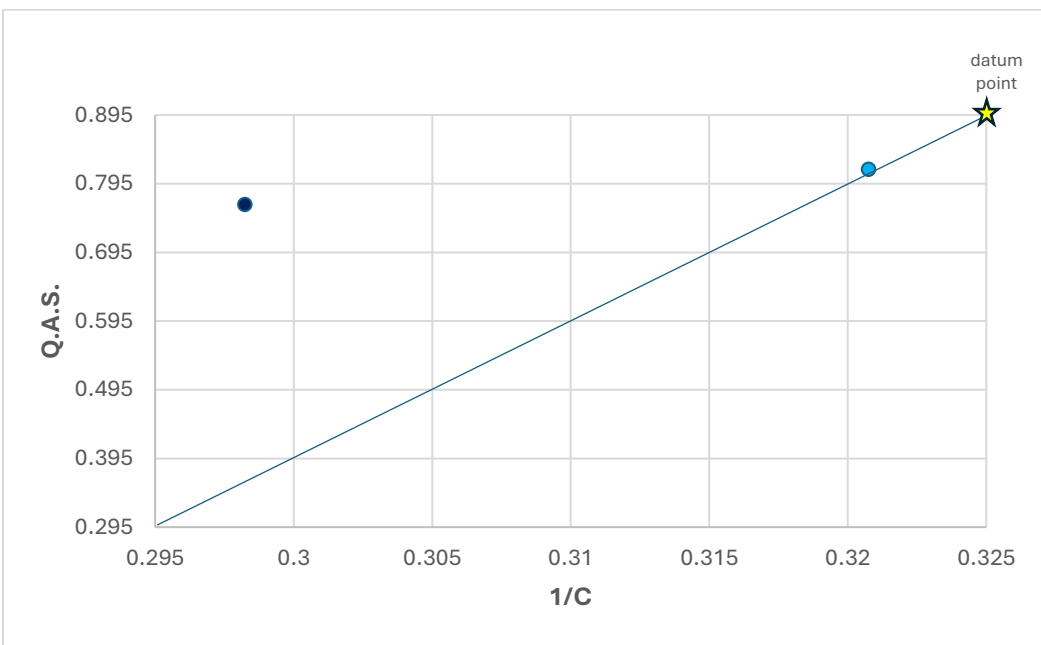
CONCEPT A	CONCEPT B	
		 

For Concept A, we decided to use a cylindrical geometry with an actuator that allows the user to occlude other channels by rotating the external surface by hand. The external surface will be designed with a specific roughness to ensure it can be easily grabbed. For Concept B, we opted for a rectangular geometry, incorporating buttons for the ingredients in a manner that makes it easy to grab and use repeatedly without effort.

2.5 Concept Evaluation

	Erg	S-R	Aes	Q	Qmax	Avg Cost	1/C	Q.A.S.
CONCEPT A	75	71	62	208	255	3.117647059	0.320754717	0.815686275
CONCEPT B	71	63	61	195	255	3.352941176	0.298245614	0.764705882

First, we computed Q as the sum of the scores assigned to some main external properties (Ergonomics, Safety, Reliability, and Aesthetics) for each actuator for both Concept A and Concept B. Next, we determined Q_{max} by summing the maximum possible scores for all actuators, assuming each property was rated at its highest possible value. By calculating the ratio of Q over Q_{max} , we obtained the Quality Average Score. For the cost calculation, we computed the average cost of the actuators and then determined the inverse of the average cost 1/C. Finally, we plotted the two quantities:



It's clear from the graph that Concept A is the best option since it's very close to the datum point (best quality and price) looking at the actuators selected for the evaluation.

3. Preliminary Lay-out

Our workflow started from the fluid selection mechanism. The element in surface design is a manifold that connects the circular section tubes for syrups and carbonated and still water with part1.

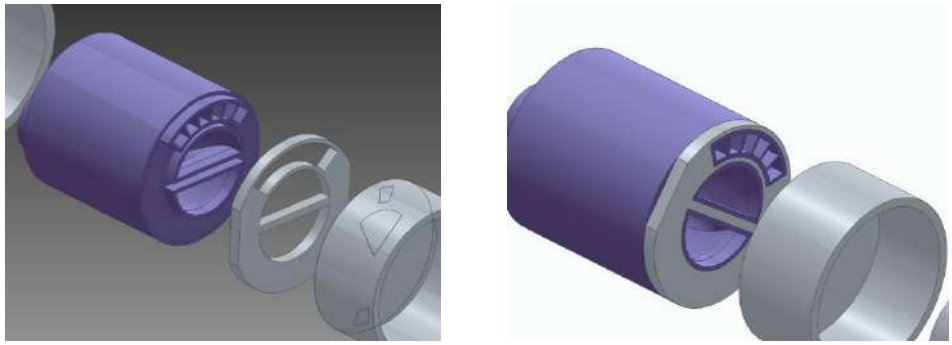


Figure 1 – Exploded view of the selecting system (surface design, part 1, part 2)

Part 2 is rotated with a finger relative to part 1, creating different situations. It is possible to select one of six syrups with still water or carbonated water by rotating 180°, and it is also possible to select only still water or only carbonated water. This actuator allows for choosing among 14 different combinations.

Figure 2 – frontal view of different selections (lemon and still water, lemon and carbonated water, carbonated water)

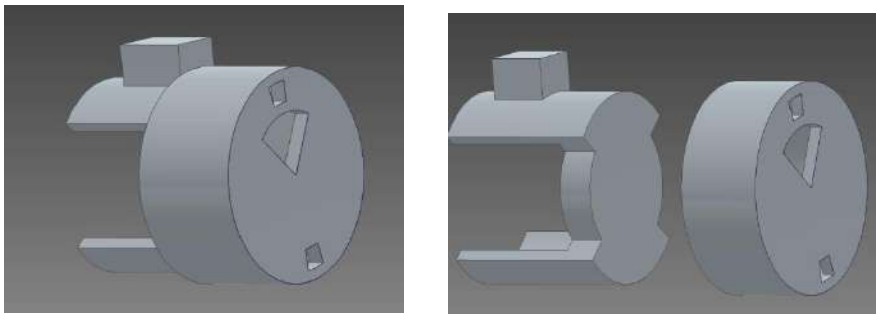


Figure 3 – Starting flow system (part 4 and part 2)

Part 4 closes the holes in part 2 thanks to a compression helical spring. By pressing the button along the axis of the soda gun, the liquid is free to flow. Releasing the button stops the flow again, so the device is always closed when not in use, preventing liquid leaks.

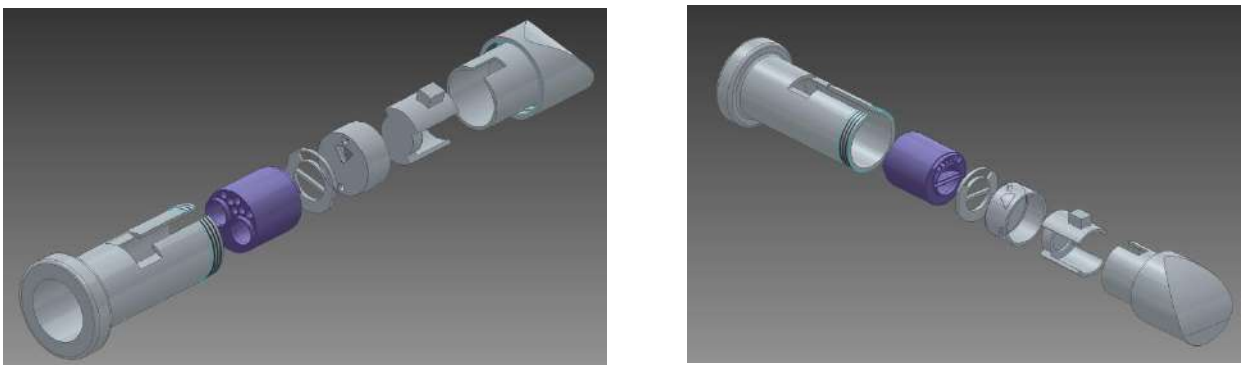


Figure 4 – Exploded view of the soda gun

Regarding the holding system, we opted for a simple solution. The support consists of a perforated plate that can be attached to any surface with screws. Attached to the plate are two inclined rods, which have a relative distance smaller than the final diameter of the soda gun but larger than the diameter of the central part. The soda gun can be placed on the fork upside down.

Figure 5 - Assembly

3.1 Surface Design

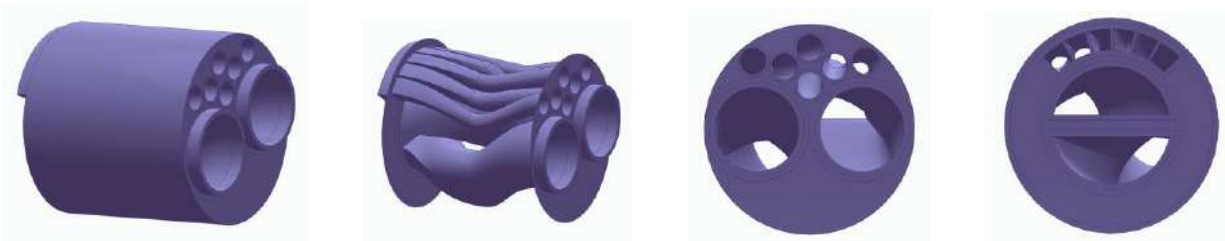


Figure 6 – Manifold in different views

We wanted to create a closed shell with surfaces constructed from closed curves. The piece designed in surface design is a manifold that must connect the standard circular section tubes for syrup and soda with the respective pouring positions on part 1, staying within the minimum section of part 3. The surfaces were created using two different methods: creating a surface from a closed curve and creating a surface by connecting two different closed curves.

The most critical part was connecting a closed curve representing a circle with a closed curve with edges. As shown by the curvature analysis, a curvature discontinuity in these internal surfaces in contact with the fluid is inevitable. For the surfaces on the side faces, orthogonality or tangency constraints were applied where we wanted the coupling with the tubes and part 1, while for the others, a curvature continuity constraint was always applied to minimize stress accumulation and to avoid areas where sugar residue might accumulate.

The rotation of the natural and sparkling water tubes is significant, which caused difficulties in preventing the surfaces from intersecting.

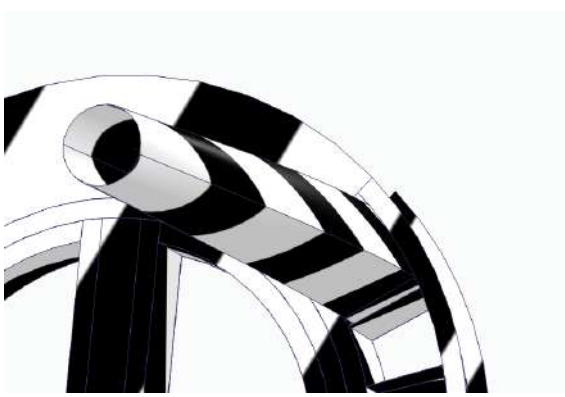


Figure 7 - Zebra Plot

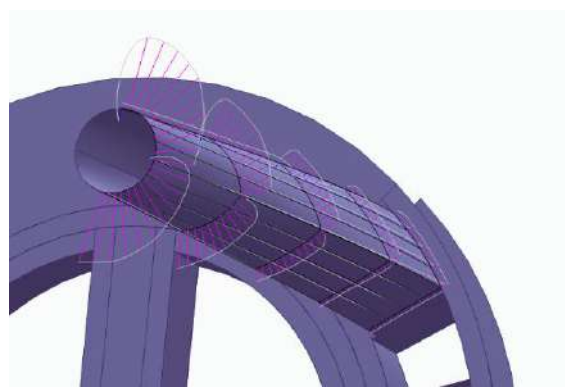


Figure 8 - Porcupine Analysis

With the surface analysis tools, we checked the surface continuity (always satisfied → closed volume) and the curvature continuity of the surfaces, which, on the other hand, is not satisfied everywhere. Sometimes, this is due to the need to connect curves with edges (rectangular sections) and curves without edges (circular sections) as shown in the figure. Other times, it was done intentionally, as in the case of the contact surfaces with the tubes.

3.2 Parametric Design

	A	B
1	height plate	160
2	width plate	100
3	plate thickness	20
4	slope	110
5	width fork	60
6	length fork	100
7	fork thickness	10
8	fork radius	24
9	fork's hole distance from plate	44
10	Hole_1_Diameter	8
11	number_holes_width	2
12	number_holes_height	2

	A	B
1	height plate	160
2	width plate	100
3	plate thickness	20
4	slope	110
5	width fork	60
6	length fork	100
7	fork thickness	10
8	fork radius	24
9	fork's hole distance from plate	44
10	Hole_1_Diameter	6
11	number_holes_width	2
12	number_holes_height	4

We assigned each dimension a value or a relation with pre-existing values so that we could quickly modify certain design elements such as the number of holes and their size. Having a component that can be modified quickly and easily allows us to have more options for the base. In the figure, we have two parts that differ in the number of holes and the size of the holes, but later we will use parametric design to size this component using the safety margin method.

3.3 Design with safety margin concept

We decided to make a study about safety margin on the basement of the soda gun, because if it goes out of service, soda gun could not be used anymore. As safety margin concept says, the study is done following 8 main reasons of safety margin variations:

1) limit about the analytic formula used to compute the equivalent condition
2) limit about the evaluation of the actual loads
3) limit about the knowledge of the probabilistic behaviour of the material strength or their acceptance tolerance
4) knowledge of the operative conditions, that means understanding if environmental conditions may impact the severity of the stress conditions or the material strength.
5) maintenance requirements and checks, since proper maintenance planning may reduce risks of failures.
6) severity of failures in the respect of safety.
7) standards/law requests.
8) weight reduction.

Each reason can be derived in a qualitative way from a scale that goes from -4 (extreme necessity of reducing X according to the ith reason) up to +4 (extreme necessity of increasing X according to the ith reason). If the sum is less than -6, safety factor is 1.15, if it is greater than -6 should be computed with a specific formula. In our case:

1) +1 (some approximations were done during FEA)
2) +2 (we didn't consider only the inertial load of soda gun, but also random loads due to frequent usage of the gun like random hits from the user)
3) +1 (a properties change on the clamped end doesn't affect its behaviour, but on the free end of the forks that hold the soda gun, a properties change could affect it)
4) -4 (environmental conditions don't impact on failures at all)
5) -1 (design and material chosen, imply very low maintenance)
6) -4 (if it fails it wouldn't affect human safety or any damages)
7) -3 (standards and laws are more than respected thanks to smooth surfaces designed)
8) -4 (clamp with 4 screws is very safe, they can hold more than 100x of the load of the basement also in dynamic (working) situations)

Since the sum is equal to -12, our final computed safety factor for the basement is X=1.15.

$$\sigma_f = \frac{M_f}{W_f} = \frac{P \cdot b}{\frac{Bh^3}{12}} = X \cdot \sigma_s \implies h = \sqrt[3]{\frac{12P \cdot b}{X \cdot \sigma_s \cdot B}} = \sqrt[3]{\frac{12 \cdot 5\,N \cdot 0.02\,m}{1.15 \cdot 44 \cdot 10^6\,Pa \cdot 0.008\,m}}$$

$$h \approx 14\,mm$$

So, the final dimensions of the basement will be h=14 mm and B=8 mm instead of h=10 mm and B=5 mm as previously.

4. Executive Design

As a company, we decided to save material. This operation can reduce costs, but it depends on the processing. We will certainly achieve an advantage in the environmental impact of the product. In this way, customers can enjoy their refreshing drink, happy to be contributing to making the world less polluted.

To reduce material, we decided to use topological optimization. This method, with an iterative approach, allows us to identify areas of the component where the material is not fully utilized. These areas can be removed by creating features. The method used is SIMP (Solid Isotropic Material with Penalization), which is a microstructure method or density-based method.

Our objective function is the compliance, depending on stresses. The stresses are evaluated under the hypothesis of linear elastic behavior of our material. The ABS, in this application, is far from the yield stress so our assumption is not incorrect.

Repeated FEAs will assign a virtual density of 1 (bulk) to some elements and 0 (void) to others, suggesting that these should be removed. The mesh was chosen to be as dense and uniform as possible, given that we had a node limit for the student version of the software used.

Initially, we decided which parts of the component should be included (design space) and which parts should be excluded from optimization (non-design space). In the non-design space, we included cylindrical volumes around the holes and the fork. The rest of the base is all design space.

Five load collectors have been defined by identifying a node that projects a constraint onto a set of associated nodes. Four are for the connections of the holes, and one is for the force applied to the fork. For the force, we assumed a load of 20 N since we do not know the total length of the tubes.

Additionally, a constraint on the maximum volume reduction was added, which we decided should not exceed 60% of the initial volume. By comparing the final STL, resulting from the topological optimization, with the FEA of each iteration, we ensured that we were sufficiently far from the material's stress limits.

In fact, as we can see from Figure 9, the maximum stress reached in the last iteration is 1.4 MPa.

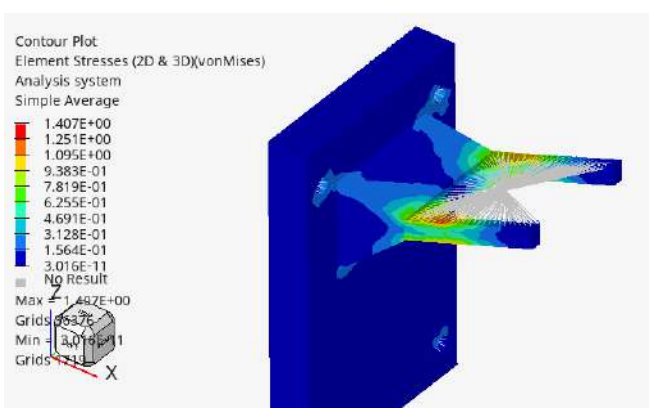


Figure 9 – FEA stress results

The optimization required 23 iterations. Below, we can observe that the initial iterations have significant differences, while the later ones tend to converge. This reassures us about the correctness of our analysis.

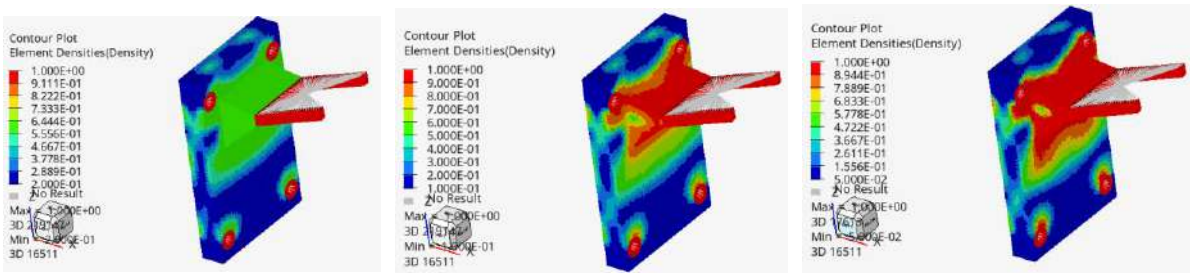


Figure 10 – First, second and third iteration

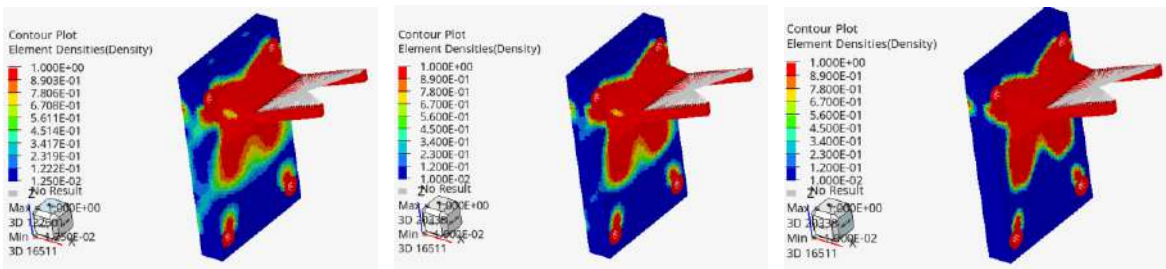


Figure 11 - 7th, 15th and 23rd iteration

We took the component at the 23rd iteration and removed 40% of the elements, thus obtaining the optimized trabecular structure. Considering this component aesthetically unacceptable, we decided to use the result to create features starting from the original model.

Finally, starting from the support model, we introduced the features that removed most of the volume according to the optimization results. We can see how the plate has disappeared to make way for a more efficient beam structure.

Subsequently, the result of the topological optimization and the proposed component.

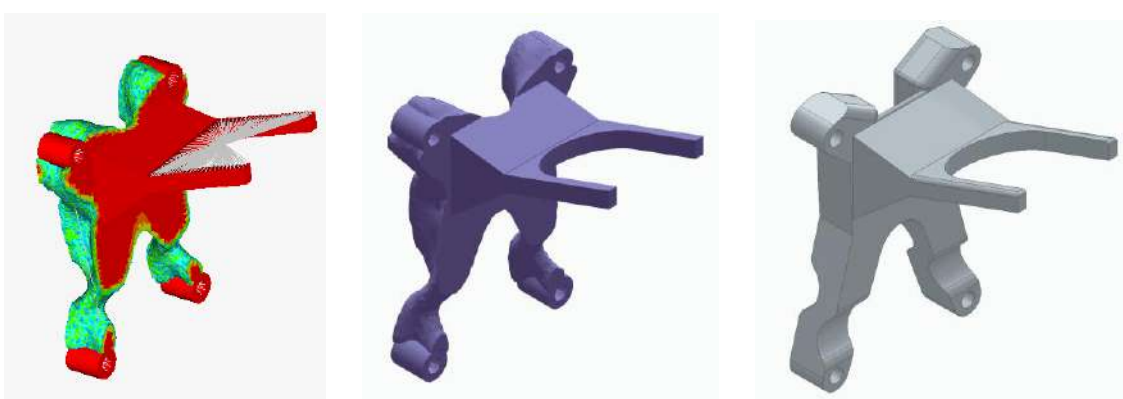


Figure 12 – Final geometry of topological optimization

However, we must note that the material savings might not be economically advantageous since we intend to produce only a few items. The initial model was much simpler and, depending on the manufacturing process, could be much less expensive to produce.